



The Challenges of Achieving Food Security in the Maghreb Region: Algeria as a Case Study

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Abstract

Food security has become one of the most pressing challenges facing the Maghreb region, where structural dependence on food imports, climate change, and demographic pressures continue to deepen the food gap. This paper explores the stakes and challenges of achieving food security in the Maghreb, with Algeria as a case study, as despite Algeria's significant agricultural potential and vast arable land, the country remains heavily reliant on imports of strategic commodities such as cereals, milk powder, and edible oils.

Since the early 2000s, the Algerian government has implemented several agricultural and rural development policies—including the National Agricultural Development Plan (PND A), irrigation expansion programs, and subsidies for key products—in an attempt to enhance domestic production and reduce vulnerability to external shocks; however, persistent structural constraints, such as water scarcity, low productivity, climate variability, and weak regional integration, continue to limit the effectiveness of these policies.

The study concludes that achieving sustainable food security in Algeria, and in the Maghreb, requires a multi-faceted approach involving the diversification of agricultural production, the adoption of climate-smart practices, the strengthening of storage and distribution systems, and the fostering of regional cooperation to optimize resource use and reduce dependence on volatile international markets.

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1. Introduction

The issue of food security assumes great importance in light of contemporary global challenges, encompassing critical economic, social, political, and security components; indeed, it is considered one of the most intractable economic problems, with its achievement being a fundamental goal for all states and international organizations, as inefficient economic policies have led to significant disparities, which are a primary reason for rising poverty and famine rates in many regions, where this situation is exacerbated by the scarcity of natural resources and their poor exploitation, which in turn causes an imbalance in environmental equilibrium—an imbalance that is directly reflected in, and leads to, food disruption. Consequently, it has become necessary to build effective plans, mechanisms, and policies to achieve self-sufficiency, particularly in the Maghreb region and in Algeria specifically.

This context gives rise to the following central research question: Have the policies and strategies of the Maghreb countries in general, and Algeria in particular, succeeded in achieving food security?

To address this problem, the study is divided into three main axes:

- Axis 1: The conceptual framework of food security.
- Axis 2: The reality and indicators of food security in the Maghreb countries.
- Axis 3: The policies adopted to achieve food security in Algeria.

First Axis: The Conceptual Framework of Food Security

Researchers widely agree that food security is a global issue whose solution lies in providing sufficient food for the world's population; it is therefore necessary to conduct an in-depth study of this problem to propose effective solutions and policies that can lead toward self-sufficiency.

First: Definition of Food Security

The concept of food security is multi-dimensional and broad, combining linguistic, jurisprudential, and legal aspects:

A/ Linguistically: It denotes tranquillity derived from the availability of foodstuffs in the required and necessary quantities. (01)

B/ Jurisprudentially: It means every citizen obtains their necessary food needs throughout the year without deprivation, whether from local production or imports. (02)

C/ Legally: The Algerian legislator defined it in Article 03 of Law 08-16 as “the obtaining and the access of every person, easily and regularly, to safe and sufficient food that allows them to enjoy an active life”. (03)

Second: The Relationship of Food Security with Similar Concepts

Food security relates to, yet is distinct from, several key concepts, which can be summarized as follows:

- **Self-sufficiency:** A policy adopted by a state to achieve sufficiency by relying on its own resources.
- **Food safety:** The extent to which food is suitable and safe for human consumption (i.e., healthy food).
- **Food gap:** The difference between local production and total demand for food, with the deficit covered through imports.
- **Food aid:** The provision of food commodities from one country to another to help meet food needs.
- **Food dependency:** Reliance on imports to meet inevitable needs.
- **Sustainable food security:** A component of sustainable agricultural development, included within the plan of sustainable economic and social development.
- **Food insecurity:** A condition in which people lack access to sufficient quantities of safe, nutritious, and healthy food. (04)

Third: Forms of Food Security

Food security can be divided into the following forms:

1/ Absolute food security (“self-sufficiency”): This occurs when a country’s production equals or exceeds local demand without resorting to imports, with any surplus exported abroad.

2/ Relative food security: This involves securing food in cooperation with others, providing foodstuffs completely, partially, or regularly.

3/ Nominal or apparent food security: This is a misleading figure, such as when a country covers 90% of its needs from domestic production while importing the rest, representing only an apparent security.

4/ Sustainable food security: This entails providing safe food and health security for present generations without compromising the rights of future generations. (05)

Fourth: Foundations and Pillars of Food Security

Achieving self-sufficiency requires states to rely on several fundamental pillars:

An abundance of food commodities for the entire population, addressing increasing demand as a preliminary stage before seeking to improve food quality.

The permanent availability of food commodities in the market, balancing quantity and quality to facilitate the move to the consumption stage.

Commodity prices being within the reach of citizens, while also considering the health dimensions of food to achieve food safety.

Food safety, which must be ensured from the stage of agricultural production through to consumption by all consumers (06), reconciling quantity with quality.

Respecting diversity and social customs, meaning that food should be acceptable to all individuals according to their needs. (07)

Fifth: Conditions for the Availability of Food Security

Specific conditions must be met to achieve food security, which are:

The availability of food.

The ability to obtain food.

The proper use of food.

Stability in the face of disruptions such as war, crises, and natural phenomena.(08)

Moreover, a fifth dimension can be added: the ability to choose. (09)

Food security also encompasses several key dimensions, which can be outlined as follows:

- **Economic dimension:** This requires a focus on supply, demand, prices, and stability in commodity markets to understand the food gap, which is caused by population increase on one hand and the irrational exploitation of natural resources on the other.

- **Political dimension:** Food is considered a strategic weapon used by resource-rich countries to exert pressure and influence national security, whether in times of peace or war; therefore, political stability is a fundamental requirement for achieving sustainable agricultural and food development.

- **Social dimension:** A relationship exists between social stability and food security—that is, between supply and demand—to avoid internal disturbances that lead to social deterioration; in brief, a balance must be struck between continuous population growth and income levels, suggesting a correlational equation whereby rising population must be matched by increasing income.

- **Cultural dimension:** This refers to the ability of a people to find ways of securing food that suit their situation and beliefs, combining their right to self-determination with their right to food.

- **Environmental dimension:** This dimension is witnessing deterioration due to a lack of coordination between agricultural-food policies and environmental protection.(10)

Sixth: Factors Affecting Food Security

The process of securing food is controlled and affected by a confluence of interrelated factors, which are:

Demographic factors: The Maghreb region faces significant pressure from a continuously growing and evolving population, a trend clearly illustrated in the accompanying table, where the total population now exceeds 109 million people, as this relentless demographic expansion places an ever-increasing demand on finite food resources, as shown in the table:

Country	Number of population (2024)
Algeria	46,814,308
Morocco	38,081,173
Tunisia	12,277,109
Libiya	7,381,023
Mauritania	5,169,395
Total	109,721,008

Table (01) showing the population for 2024–2025, based on World Bank statistics.

B. Natural factors: A critical challenge is the ongoing decline in the proportion of arable land, a situation prevalent across all Maghreb countries, including Algeria. This is primarily due to the adverse effects of climate change, which manifest through an increasing frequency of droughts and the relentless advance of desertification. (11)

C. Technological factors: The region suffers from a significant lack, if not a complete absence, of modern biotechnology and advanced agricultural techniques; such technologies are essential for the efficient investment in and management of precious resources like water and agricultural soil, and their

deficiency directly results in the declining productivity of goods and crops.

D. Political factors: Political stability constitutes an absolute necessity and a prerequisite for ensuring the consistent availability of food in quantities sufficient to match population growth, as instability disrupts supply chains, hinders long-term agricultural planning, and jeopardizes food distribution systems. (12)

Axis Two: The Reality and Indicators of Food Security in the Maghreb Region

The Maghreb countries collectively experience a high and structurally embedded dependence on food imports, particularly for cereals; Algeria, Morocco, and Tunisia, for instance, import between 50% and 70% of their wheat needs, while Libya relies almost entirely on foreign markets due to the pronounced weakness of its domestic agricultural sector; while Mauritania, despite its ongoing attempts to promote rice cultivation in the riverbank areas, it remains unable to meet burgeoning domestic demand.

This import dependency is influenced by a complex interplay of several critical factors, including but not limited to:

1. Food Commodity Prices: Given that the Maghreb countries import a significant portion of their staple foods—such as cereals, oils, sugar, and animal products—from the global market, fluctuations in international commodity prices directly and acutely affect both national economies and citizens' ability to access these products in sufficient quantity and quality.

2. Income Levels of Individuals: The economic capacity to secure food varies considerably across the region; in 2023, the average per capita share of GDP ranged from a modest 1,106 USD to 5,348 USD, with averages exceeding 4,000 USD in both Algeria and Tunisia, while Mauritania and Morocco reported figures below 3,000

USD, highlighting disparities in purchasing power.

3. Food Price Index: Recent global price trends show a decrease for cereals, sugar, and vegetable oils, yet a concurrent increase for meat and dairy products; this divergent movement necessitates that Maghreb countries strategically intensify domestic production of those goods experiencing price inflation to mitigate import costs.

4. Per Capita Consumption of Calories, Protein, and Fat: The nutritional intake of the average Maghreb citizen is estimated at approximately 3,254 kilocalories per day, complemented by 90 grams of protein and 80 grams of fat, figures which, while seemingly adequate on aggregate, often mask significant inequalities in distribution and access. (12)

To conduct a rigorous analysis of the state of food security in the Maghreb, it is essential to rely on specific indicators that measure the degree of self-sufficiency and the extent of the food gap:

1. Self-Sufficiency Ratio of Food Commodities:

During the 1990s, Maghreb countries suffered from chronically low self-sufficiency ratios for many essential food commodities; for cereals, the rate reached only 41%, meaning a staggering 59% were imported, a situation particularly evident in Algeria and Mauritania. These rates can be categorically divided into three distinct groups:

1.1. Groups with high self-sufficiency rates: This category includes potatoes, vegetables, and fruits, which collectively achieved a rate of 36.2%.

1.2. Groups with medium self-sufficiency rates: This encompasses red meat, dairy products, poultry meat, and legumes, with an average rate of 54.26%.

1.3. Groups with low self-sufficiency rates: This critical group, which includes

the foundational staples of cereals, oils, and sugar, languished at a mere 30.82%.

2. Food Trade Balance Indicator:

Despite numerous reforms within the agricultural sector, the trade balance for these countries consistently registers a deficit; it is therefore imperative to identify systemic weaknesses for remediation and strengths for reinforcement, as cultivating a resistant economy is a primary defense against foreign dominance over essential food needs. This deficit manifests in two groupings:

2.1. First group: Comprising Libya, Algeria, and Tunisia, whose food exports covered only 38.5% of their import needs, leaving a substantial gap of 60%, a direct consequence of their heavy reliance on foreign markets.

2.2. Second group: Including Morocco and Mauritania, which managed to achieve a more favourable balance, with export revenues covering a significantly larger portion of their food import requirements.

3. Global Hunger Index (GHI) in Maghreb Countries:

The Global Hunger Index (GHI), which assesses severity based on criteria of undernourishment, child mortality, and child underweight data, reveals varied levels of food insecurity across the region:

- Algeria scored less than 5 on the index, a serious level also seen in Morocco.
- Mauritania registered a score of 22.7, which is classified as a moderate level of hunger.
- Libya and Tunisia were classified in the alarming category.

A critical conclusion to be drawn is that the food security of the Maghreb citizen remains largely in foreign hands, granted or withheld according to the will and under the monopolistic conditions of exporting countries, which ultimately serve their own geopolitical and economic interests. (13)

Axis Three: Policies to Achieve Food Security in Algeria

An examination of the principal policies and strategies employed by the Algerian government to achieve food security must first acknowledge the complex interplay of factors that have historically impeded progress, constituting the fundamental causes of the nation's persistent food deficit.

First: Reasons for Not Achieving Food Security in Algeria

The impediments are usefully categorized into internal (national) and external (international) factors, which collectively constrain agricultural output and stability:

A. The Role of Internal Factors:

- **A legacy of inconsistent agricultural policies**, commencing with the self-management policy in 1963, followed by the Agrarian Revolution in 1972 under the principle of “the land belongs to the one who cultivates it,” and subsequently the restructuring phase from 1981 to 1990; the following decade saw the agricultural sector neglected amidst a severe economic crisis precipitated by falling oil prices and concomitant political instability, a period also marked by the implementation of privatization and structural adjustment policies, alongside the Economic and Agricultural Recovery Program which aimed, albeit with limited success, to increase production and reduce rural-to-urban migration.
- **A profound demographic-pressure imbalance**, with the population expanding from 7 million post-independence to 45 million in 2022—a figure projected to reach 50 million by 2030—

thereby vastly outpacing the growth of domestic food production.

- **An entrenched economic over-reliance on hydrocarbon revenues**, which has historically diverted focus and investment away from productive sectors like agriculture.
- **The persistent fragmentation of viable agricultural land** due to complex issues of ownership distribution and inheritance laws.
- **Significant urban encroachment**, whereby buildings, projects, and infrastructure increasingly consume fertile lands, particularly in the more productive northern regions.
- **The continued prevalence of traditional production methods** within the agricultural sector, limiting yields and efficiency.
- **The escalating impacts of climate change and global warming**, exacerbating environmental challenges such as desertification, drought, pollution, and reduced rainfall, all of which directly contribute to lower land productivity.

B. The Role of External Factors:

- **Heightened vulnerability to global food price volatility**, driven by increasing world population and production declines in exporter nations due to adverse climate conditions.
- **Disruption from international geopolitical crises**, most notably the Russia-Ukraine war's severe impact on global grain and corn supplies—a critical concern for Algeria, given that approximately 270 million people, including its own population, depend on imports from this region, affecting stability across the Middle East, Central Asia, and Africa.

- **The cascading effects of the COVID-19 pandemic**, which disrupted global supply chains and spurred illegal speculation in food trade within Algeria, compelling authorities to enact strict deterrent laws to stabilize markets. (14)
- **The strategic use of food as a geopolitical instrument by developed nations**, creating a form of dependency wherein non-producing countries face political pressure and blackmail; for instance, Algeria's need to import 50% of its dry grains leaves it exposed to price shocks and supply manipulations in major markets like Chicago and Paris, a vulnerability compounded by tense relations with key trade partners like France or the European Union.

Second: Policies and Strategies to Achieve Food Security in Algeria

In response to these challenges, the Algerian government has formulated a multi-pronged strategy encompassing both national initiatives and international engagement:

- Providing direct support and incentives to farmers, producers, and agricultural cooperatives to expand output of heavily imported staples, a core objective of the National Agricultural and Rural Development Plan (PNDAR), which seeks to modernize farming and enhance productivity.
- Channeling investment into the High Plateaus regions and facilitating financing for agricultural activities, with desert farming promoted as a strategic future for large-scale cultivation of cereals, potatoes, fodder, and palm trees.
- Bolstering agro-industries through medium- and long-term loans from local banks, aiming to reduce dependency on imported semi-processed goods by adding value locally—transforming grains into pasta, tomatoes into canned goods, and milk into cheese and other derivatives.
- intensifying public investment in water resource management via dam construction (approximately 80 are operational) and the adoption of modern irrigation techniques like pivot and drip systems, all guided by the comprehensive 2030 Water Strategy that includes seawater desalination and wastewater reuse.
- Actively pursuing international partnership agreements to secure technology transfer and investment; notably, in 2024, Algeria established three significant partnerships:
 - **The Algerian–Qatari Partnership (“Baladna Algeria Project”)**: a large-scale, \$3.5 billion investment with the Qatari company Baladna for powdered milk production, encompassing integrated fodder cultivation, livestock raising, and processing on 11,700 hectares, aiming to cover 50% of national demand and create 5,000 jobs, alongside cooperation within the Qatar-founded Global Drylands Alliance (GDA).
 - **The Algerian–Italian Partnership**: a €420 million "Mega Agricultural Project" in Timimoun (2024-2028) focused on cereals and legumes, facilitated by Italy's "Maitey Plan" for technology transfer, including advanced satellite monitoring for crop and water management.
 - **The Algerian–Saudi Partnership**: an expansion of the "Ikhwan Farm Project" in El Menea to 20,000 hectares with an investment of

nearly 12 billion Algerian dinars, dedicated to strategic fodder, cereal farming, and cattle raising, and providing substantial employment. (15)

These international partnerships constitute major strategic axes for enhancing food security through capital investment and land development; nevertheless, they encounter serious environmental challenges—particularly groundwater depletion in arid regions—rendering them a necessary but insufficient component of a comprehensive solution.

Conclusion:

It is unequivocally clear from the preceding analysis that the Maghreb region is endowed with ample natural, hydrological, human, and technological resources, as well as the practical experience, to achieve food security; realizing this potential, however, is strictly contingent upon the implementation of coherent strategies to protect and develop these resources, minimize waste, ensure their rational utilization, and deploy them with optimal efficiency.

Notwithstanding formidable internal and external challenges, it is absolutely imperative for Maghreb nations to craft a highly effective and deeply integrated strategy; this necessitates reinforcing regional agricultural cooperation, harmonizing development plans, facilitating the seamless movement of goods within the Maghreb, and establishing a regional market shielded from global fluctuations through coordinated policies on customs protection and price stabilization.

Therefore, the achievement of sustainable food security in Algeria remains intrinsically dependent on profound and enduring reforms within agricultural policy, a dedicated embrace of innovation and technology, and the strategic cultivation of more resilient regional and international partnerships. Ultimately, food security transcends mere economic policy to become a strategic imperative essential for safeguarding Algeria's national sovereignty and fortifying its capacity to navigate future uncertainties.

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